Plasma adiponectin concentrations are associated with body composition and plant-based dietary factors in female twins.

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Circulating adiponectin is emerging as an important link between obesity, type 2 diabetes, and cardiovascular disease (CVD). However, the spectrum of lifestyle factors that modulate the adiponectin concentration remains to be elucidated, particularly among women. We conducted a cross-sectional study of 877 female twin pairs from the TwinsUK adult twin registry. Using a co-twin design, we examined dietary and body composition influences on adiponectin by conducting matched, within-pair analyses to eliminate confounding. Following multivariable adjustment within-twin pairs, significant influences on adiponectin (log-transformed, percent change per SD of the dietary/body composition variable) were observed for nonstarch polysaccharides (3.25%; 95% CI: 0.06, 6.54; $P < 0.05$) and magnesium intake (3.80%; 95%CI: 0.17, 7.57; $P < 0.05$), with a trend toward an association for fruit and vegetable (F&V) intakes (2.55%; 95% CI: -0.26, 5.45; $P = 0.08$). These modest positive associations cannot be explained by confounding through other lifestyle factors shared by the twins. A significant relationship between adiponectin and 3 derived dietary patterns (F&V, dieting, traditional English), carbohydrate, protein, trans fat, and alcohol intake was also observed. Strong inverse associations with adiponectin were observed for BMI (-10.72%; 95% CI: -13.78, -7.55), total (-6.89%: 95% CI: -10.34, -3.30; $P < 0.05$), and central fat mass (-12.50%; 95% CI: -15.82, -9.05; $P < 0.05$); these relationships were significant both when twins were analyzed as individuals and when characteristics were contrasted within-twin pairs, suggesting a direct effect. We observed modest associations between dietary factors and adiponectin in female twins, independent of adiposity, and report strong inverse associations with body composition. These data reinforce the importance of weight maintenance and increasing consumption of diets rich in plant-based foods to prevent CVD and type 2 diabetes.